



preset VR1 so as to get a voltage of 1.2 volts (with battery voltage at 10.5V).

Initially, when the battery is fully charged, the voltage at the inverting

input of IC1 is higher than the non-inverting input and output pin 12 of IC1 remains low. The reset pin (pin 4) of IC2 connected to pin 12 of IC1 also re-

mains low and the astable multivibrator built around IC2 does not oscillate. As a result, LED1 does not flicker.

When the battery voltage falls below 10.5V, the voltage at the inverting input of IC1 becomes lower than the non-inverting input and the output of IC1 goes high. The reset pin of IC2 connected to pin 12 of IC1 also goes high and the astable multivibrator built around IC2 starts oscillating. LED1 flickers to indicate that the battery voltage is low and the battery needs to be charged before further use. Both IC1 and IC2 operate off regulated +5V DC generated by voltage regulator IC 7805 (IC3).

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Mount LED1 and switch S1 on the front side of the case. Connect a 12V battery to check its voltage level. ●

NUMERIC WATER-LEVEL INDICATOR

■ DANIYAL SYED

Most water-level indicators for water tanks are based upon the number of LEDs that glow to indicate the corresponding level of water in the container. Here we present a digital version of the water-level indicator. It uses a 7-seg-

ment display to show the water level in numeric form from '0' to '9'.

The circuit works off 5V regulated power supply. It is built around priority encoder IC 74HC147 (IC1), BCD-to-7-segment decoder IC CD4511 (IC2), 7-segment display LTSS43 (DIS1) and a few discrete components. Due to high input impedance, IC1 senses water in



the container from its nine input terminals. The inputs are connected to +5V via 560-kilo-ohm resistors. The ground terminal of the sensor must be kept at

